## Exercise – 10: Implement a Case on Network Forensics

**Aim**

To implement a **network forensic investigation** by capturing and analyzing network traffic to detect suspicious activities.

**Objectives**

- To understand the concept of **Network Forensics**.
- To capture network packets from a live network.
- To analyze suspicious traffic and detect possible cyber attacks.
- To learn how forensic tools are used in network investigations.

**Software Requirements**

- Kali Linux / Ubuntu / Windows
- Network forensic tools:
    - Wireshark (Packet Analyzer)
    - tcpdump (Command-line packet capture tool)
    - Network connection

**Hardware Requirements**

- Computer system
- Network connection (LAN / WiFi)

**Theory**

**Network Forensics**

Network Forensics is the process of **capturing, recording, and analyzing network traffic** to investigate cyber attacks or unauthorized activities.

It helps investigators determine:

- Source of network attacks
- Suspicious data transfers
- Unauthorized access attempts
- Malware communication

Network forensic analysis is widely used in:

- Cybercrime investigations
- Intrusion detection
- Incident response
- Network security monitoring

**Network Forensic Investigation Process**

1. Packet Capture
2. Traffic Monitoring
3. Traffic Analysis
4. Identifying Suspicious Activity

5. Evidence Documentation

**Tool Used: Wireshark**
**About Wireshark**
Wireshark is a **network protocol analyzer** used to capture and inspect packets traveling across a network.
Wireshark allows investigators to analyze:

- HTTP traffic
- DNS queries
- TCP connections
- Network protocols

**Procedure**
**Step 1: Install Wireshark**

1. Download Wireshark from official website.
2. Install the software.
3. Open the application.

**Step 2: Select Network Interface**

1. When Wireshark starts, it displays available interfaces.
2. Choose the active interface:
    - WiFi
    - Ethernet
3. Double-click the interface to start capturing packets.

**Step 3: Capture Network Traffic**
Once capture starts, Wireshark displays live packets including:

- Source IP address
- Destination IP address
- Protocol type
- Packet length

Packets will continuously appear in the capture window.

**Step 4: Apply Filters**
Wireshark filters help investigators analyze specific traffic.
**HTTP traffic filter**
http
**DNS traffic filter**
dns
**IP address filter**
ip.addr == 192.168.1.1
Filters help investigators focus on suspicious activity.

**Step 5: Analyze Packets**
Click on a packet to view detailed information:

- Packet headers
- Source and destination addresses
- Protocol information
- Payload data

Investigators analyze packets to identify suspicious patterns.

**Using tcpdump (Linux)**
tcpdump is a **command-line network packet analyzer**.

**Capture packets**
tcpdump -i eth0
This command captures packets from interface **eth0**.

**Save captured packets**
tcpdump -i eth0 -w capture.pcap
This saves packets to a file for later analysis.

**Read captured file**
tcpdump -r capture.pcap
This displays packets from the saved capture file.

**Example Case Study**
**Scenario**
A system administrator suspects that a system is communicating with an unknown external server.
**Investigation Steps**

1. Capture network traffic using Wireshark.
2. Apply filters to identify suspicious IP addresses.
3. Analyze unusual traffic patterns.
4. Examine protocols used for communication.
5. Document evidence.

**Findings**

- Suspicious external IP detected.
- Unusual data transfer observed.
- Possible malware communication identified.

**Observation**

| Activity | Evidence Found |
|---|---|
| Packet capture | Multiple network packets captured |
| DNS queries | Domain name lookups detected |
| HTTP traffic | Web browsing activity identified |

| | |
|---|---|
| Suspicious IP | External server communication |

**Applications**

- Cybercrime investigation
- Intrusion detection
- Malware network analysis
- Network monitoring

**Precautions**

- Perform packet capture only on **authorized networks**.
- Maintain confidentiality of captured data.
- Document forensic evidence carefully.

**Result**

- Thus, network traffic was captured and analyzed using forensic tools. Suspicious network activity was identified through packet analysis.

**Viva Questions**

1. What is network forensics?
2. What is packet capture?
3. What is Wireshark used for?
4. What is a network packet?
5. What is the purpose of filters in Wireshark?
6. What is tcpdump?